

MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 13: d-Separation, Good and Bad Controls

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Stanford University

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Readings: *Applied Causal Inference Powered by ML and AI*, §7, §11.

Recap: What We Learned in Lecture 12

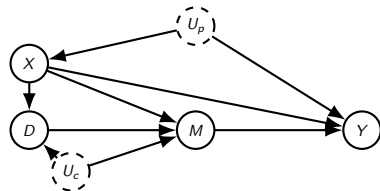
Last lecture we introduced a powerful new framework:

1. **DAGs from domain knowledge:** We translated human-interpretable assumptions about the hospital messaging application into a **Directed Acyclic Graph (DAG)**.
2. **Non-parametric SEMs:** The DAG corresponds to a **structural equation model** — a set of equations $V_i = f_i(\text{Pa}(V_i), \varepsilon_i)$ with independent noise terms.
3. **Interventions:** We formalized $\text{fix}(D = d)$ by replacing D with d in all downstream equations, giving an **interventional interpretation** to potential outcomes.
4. **SWIGs:** The Single World Intervention Graph depicts the interventional world as a Bayesian network, enabling us to **deduce** conditional exogeneity.

Recap: The Hospital Messaging Case Study

Variables:

- ▶ Y = glucose time-in-range
- ▶ D = whether doctor sent a message
- ▶ M = text of the message
- ▶ X = patient health data
- ▶ U_p = patient's latent health state
- ▶ U_c = doctor's workload/capacity

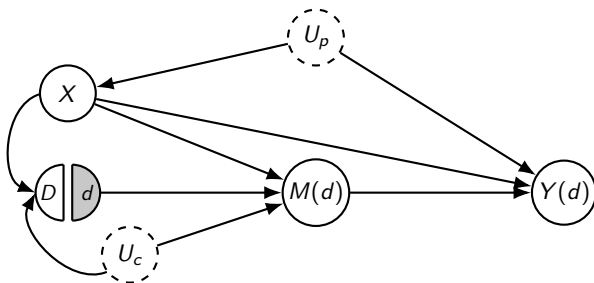


The Key Question

Can we **deduce** conditional exogeneity $Y(d) \perp\!\!\!\perp D \mid X$ from this DAG?

Recap: The SWIG for $\text{fix}(D = d)$

We construct the SWIG by splitting D into two halves:



What we need to check

In the SWIG, how can we visually check whether $Y(d) \perp\!\!\!\perp D \mid X$???

How do we check conditional independence in a Bayesian network?

Part I: d-Separation

The Graphical Criterion for Conditional Independence

d-Separation: The Graphical Independence Criterion

Definition: d-Separation

Two nodes A and B are **d-separated** given a set $\mathbf{S}$ if **every** undirected path between A and B is **blocked**. A path is blocked if it contains:

1. A **chain** $\cdots \rightarrow W \rightarrow \cdots$ or **fork** $\cdots \leftarrow W \rightarrow \cdots$ where $W \in \mathbf{S}$, or
2. A **collider** $\cdots \rightarrow W \leftarrow \cdots$ where $W \notin \mathbf{S}$ and no descendant of W is in $\mathbf{S}$.

Key Property

In a Bayesian network, if A and B are d-separated given $\mathbf{S}$, then $A \perp\!\!\!\perp B \mid \mathbf{S}$.

Strategy: To check $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG, enumerate all paths between D and $Y(d)$ and check whether each is blocked by conditioning on X .

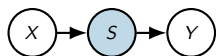
Intuition for d-Separation: The Three Building Blocks

Every path in a DAG is a sequence of three patterns. Three variables X , S , Y :

Intuition for d-Separation: The Three Building Blocks

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Chain (Mediator):



$p(x, y \mid s)$:

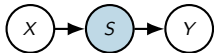
$$\begin{aligned} &= \frac{p(x)p(s \mid x)p(y \mid s)}{p(s)} \\ &= \underbrace{f(x, s)}_{\frac{p(x)p(s|x)}{p(s)}} \cdot \underbrace{g(y, s)}_{p(y|s)} \end{aligned}$$

Blocked ✓

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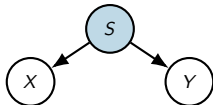


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Blocked ✓

Fork (Common Cause):



$p(x, y \mid s)$:

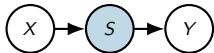
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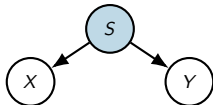


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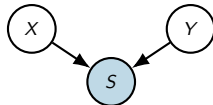


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Blocked ✓

Collider (Common Effect):



$p(x, y | s)$:

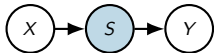
$$\begin{aligned} &= \frac{p(x)p(y)p(s | x, y)}{p(s)} \\ &\neq f(x, s) \cdot g(y, s) \end{aligned}$$

Opened! ✗

Intuition for d-Separation: The Three Building Blocks

Every path in a DAG is a sequence of three patterns. Three variables X , S , Y :

Chain (Mediator):

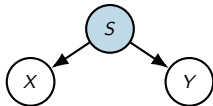


$$p(x, y | s):$$

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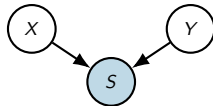


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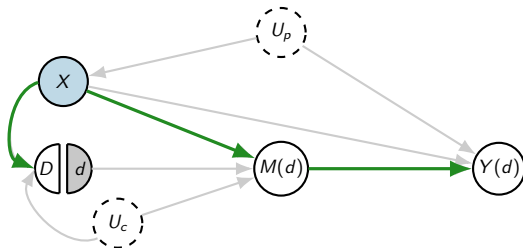
d-Separation generalizes these three patterns to arbitrary graphs: trace every path, check each node as chain, fork, or collider relative to **S**.

Part II: Applying d-Separation

Checking Conditional Exogeneity in the Hospital Example

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:

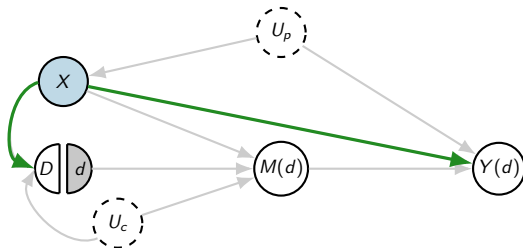


Path 1: $D \leftarrow X \rightarrow M(d) \rightarrow Y(d)$

Blocked — fork at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:

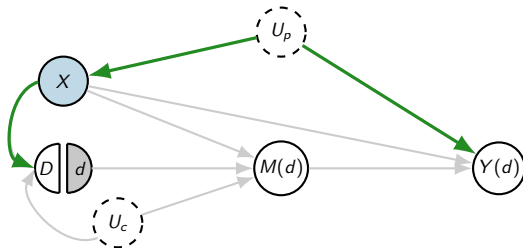


Path 2: $D \leftarrow X \rightarrow Y(d)$

Blocked — fork at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

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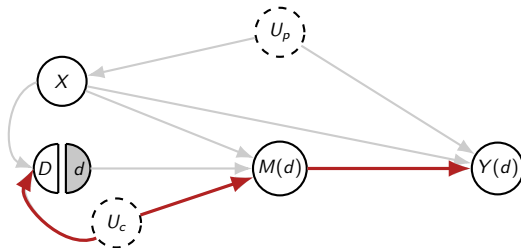


Path 3: $D \leftarrow X \leftarrow U_p \rightarrow Y(d)$

Blocked — chain at $X \in \mathbf{S}$

Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:

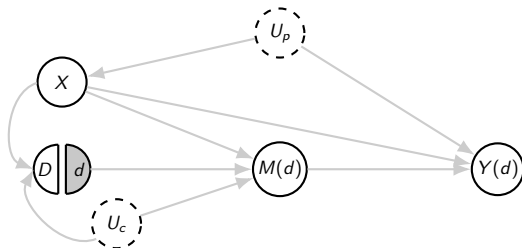


Path 4: $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$

Open! — U_c unobserved, X not on path

Checking Paths: $Y(d) \not\perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between D and $Y(d)$:



Conclusion

$Y(d) \not\perp\!\!\!\perp D \mid X$. Conditional exogeneity **fails!** The open path $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ cannot be blocked because U_c is **unobserved**.

Pause and Reflect

We cannot conclude that:

$$Y(d) \perp\!\!\!\perp D \mid X$$

If you cannot verify it on the SWIG, you should not use it for identification!

Why Does This Fail? — The Intuition

The Subtle Confounding Through the Mediator

The SWIG reveals that intervening on D alone does **not** remove all confounding:

- ▶ $\text{fix}(D = 1)$ forces every doctor to send a message, but it does **not** control *what kind* of message is sent.
- ▶ U_c (doctor's workload/capacity) still influences the **quality** of the message $M(d)$.
- ▶ So the path $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ remains **open**: the left half of D inherits U_c , and U_c flows through $M(d)$ to $Y(d)$.

In plain language:

- ▶ In the observed data, doctors who *chose* to send messages typically had **more time**, so they crafted **thoughtful, effective** messages.
- ▶ Under $\text{fix}(D = 1)$, overloaded doctors are *forced* to send messages too — but theirs would be **rushed and generic**.

Why Does This Fail? — Implications

- ▶ Naïvely comparing “message sent” vs. “no message” in the data **overestimates** the causal effect of $\text{fix}(D = 1)$.
- ▶ The observed messages are **systematically better** than the ones we would get under the intervention, because doctors who chose to send them had more capacity.

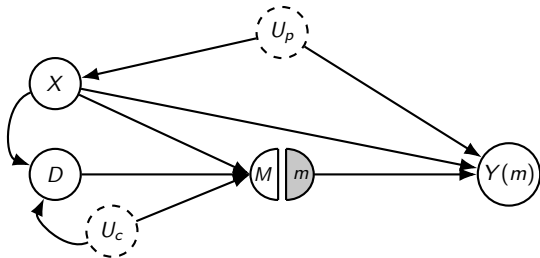
What Should We Do Instead?

- ▶ **Option 1:** Condition on U_c (physician workload) — removes the confounding by controlling for capacity.
- ▶ **Option 2:** Intervene on M directly ($\text{fix}(M = m)$) — standardize the message content, bypassing the confounding entirely.

A Better Intervention: $\text{fix}(M = m)$

The SWIG told us the problem: U_c flows through $M(d)$ to $Y(d)$.

Idea: Instead of intervening on D , intervene directly on the **message content** M , bypassing the confounding entirely.

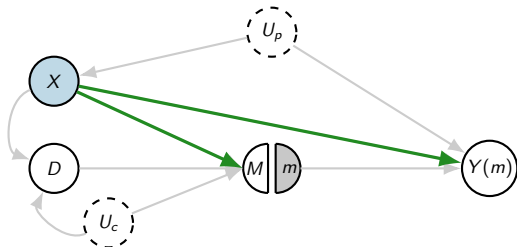


What we need to check

$$Y(m) \perp\!\!\!\perp M \mid X \quad ???$$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:

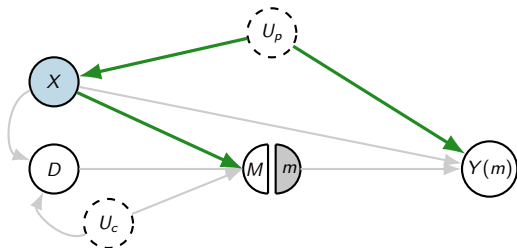


Path 1: $M \leftarrow X \rightarrow Y(m)$

Blocked — fork at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:

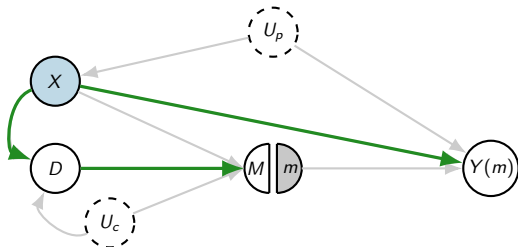


Path 2: $M \leftarrow X \leftarrow U_p \rightarrow Y(m)$

Blocked — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:

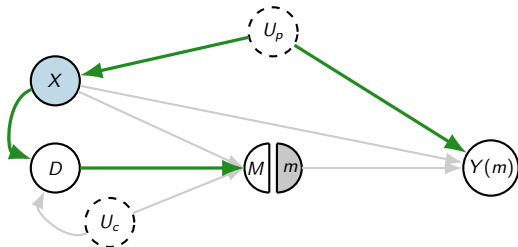


Path 3: $M \leftarrow D \leftarrow X \rightarrow Y(m)$

Blocked — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

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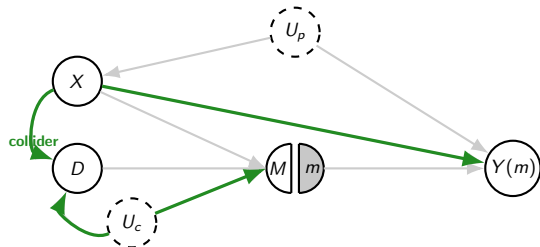


Path 4: $M \leftarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$

Blocked — chain at $X \in \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:

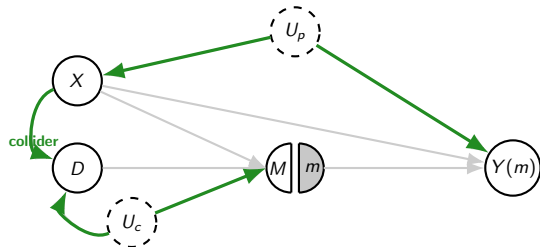


Path 5: $M \leftarrow U_c \rightarrow D \leftarrow X \rightarrow Y(m)$

Blocked — collider at D , $D \notin \mathbf{S}$

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:

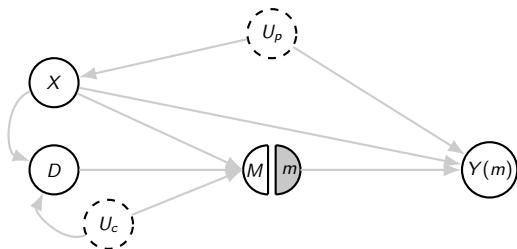


Path 6: $M \leftarrow U_c \rightarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$

Blocked — collider at D

Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on $\mathbf{S} = \{X\}$. We highlight each path between M and $Y(m)$:



All Paths Blocked!

Every path from M to $Y(m)$ is blocked by X . Paths through U_c hit the **collider** at D ($U_c \rightarrow D \leftarrow X$), which blocks them without conditioning.

Conditional Exogeneity **Holds** for $\text{fix}(M = m)$!

Result

$$Y(m) \perp\!\!\!\perp M \mid X \quad \checkmark$$

Conditional exogeneity holds when we intervene on the **message** M . We can apply all identification and estimation strategies from prior lectures to estimate the causal effect of the message content!

Voilà! We have deduced conditional exogeneity as a **conclusion**, where the only assumption was the **human-interpretable DAG**.

*But note: we could **not** have deduced this for the binary decision D alone — the SWIG revealed the subtle confounding through the mediator!*

Part III: Examples

Good and Bad Controls — A Gallery of Examples

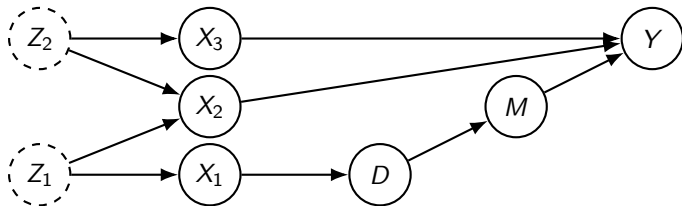
Example 1: Drug Efficacy with Mediation — Scenario

Scenario: A pharmaceutical company wants to estimate the effect of a **drug** (D) on **recovery** (Y).

- ▶ D = whether the patient receives the drug (binary treatment)
- ▶ Y = patient recovery outcome (e.g., symptom score after 6 months)
- ▶ M = **biomarker level** — the drug changes this biomarker, which affects recovery ($D \rightarrow M \rightarrow Y$)
- ▶ X_1 = patient's **insurance type** — better insurance $\Rightarrow$ more likely to be prescribed the drug
- ▶ X_2 = patient's **disease severity at baseline** (comorbidity index) — affects recovery directly
- ▶ X_3 = patient's **age / age-related decline** — older patients have worse recovery outcomes
- ▶ Z_1 = unobserved **employer type** — physically demanding jobs lead to worse health (X_2) and different insurance plans (X_1)
- ▶ Z_2 = unobserved **family history of disease** — genetic predisposition affects both disease severity (X_2) and age / age-related decline (X_3)

Task (5 min): Given the DAG on the next slide, what are valid adjustment sets for the causal effect of D on Y ?

Example 1: Drug Efficacy with Mediation — DAG



Task: What are **valid adjustment sets** (i.e., sets of variables to control for) for the causal effect of D on Y ?

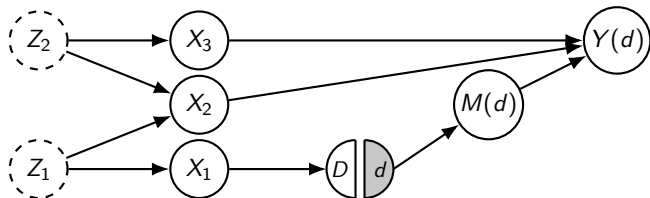
Poll: Valid Adjustment Sets

Poll Question

In the Example 1 DAG, what are the valid adjustment sets

- (a) The emptyset
- (b) $\{X_1\}$
- (c) $\{X_2\}$
- (d) $\{X_3\}$
- (e) $\{X_1, X_2\}$
- (f) $\{X_2, X_3\}$
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- (h) $\{X_1, X_2, X_3\}$

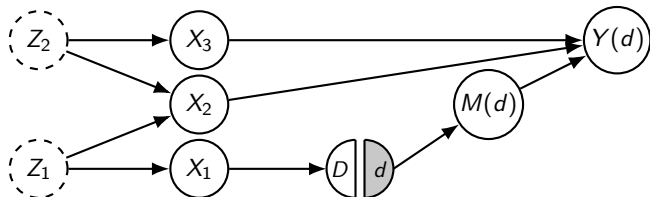
Example 1: Drug Efficacy — SWIG for $\text{fix}(D = d)$



Question

Using d-separation on this SWIG, determine: does $Y(d) \perp\!\!\!\perp D$ hold unconditionally? If not, what is a valid conditioning set $\mathbf{S}$ such that $Y(d) \perp\!\!\!\perp D \mid \mathbf{S}$? List all non-causal paths between D and $Y(d)$.

Example 1: Effect on Mediator



Question

Using d-separation on this SWIG, determine: does $M(d) \perp\!\!\!\perp D$ hold unconditionally? If not, what is a valid conditioning set $\mathbf{S}$ such that $M(d) \perp\!\!\!\perp D \mid \mathbf{S}$?

Poll: Valid Adjustment Sets

Poll Question

In the Example 1 DAG, when estimating the effect of D on the mediator M , what are the valid adjustment sets

- (a) The emptyset
- (b) $\{X_1\}$
- (c) $\{X_2\}$
- (d) $\{X_3\}$
- (e) $\{X_1, X_2\}$
- (f) $\{X_2, X_3\}$
- (g) $\{X_1, X_3\}$
- (h) $\{X_1, X_2, X_3\}$

Example 2: Drug Efficacy with Confounding — Scenario

Scenario: Same pharmaceutical setting as Example 1, but now **disease severity** (X_2) also directly influences treatment assignment.

- ▶ Sicker patients (higher X_2) are *more likely* to be prescribed the drug *and* have worse recovery outcomes.
- ▶ This creates a **confounding path**: $D \leftarrow X_2 \rightarrow Y$.
- ▶ All other variables remain the same as in Example 1.

Task (5 min): With the additional edge $X_2 \rightarrow D$, what are valid adjustment sets now?

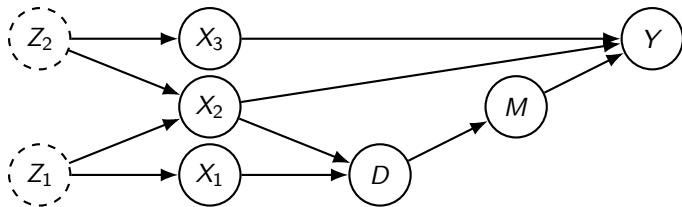
Poll: Valid Adjustment Sets

Poll Question

In the Example 2 DAG, what are the valid adjustment sets

- (a) The emptyset
- (b) $\{X_1\}$
- (c) $\{X_2\}$
- (d) $\{X_3\}$
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- (f) $\{X_2, X_3\}$
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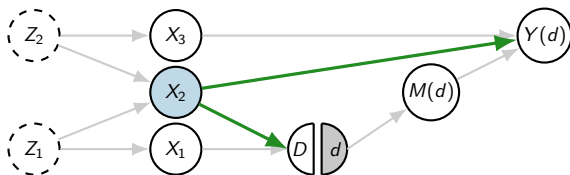
Example 2: Drug Efficacy with Confounding — DAG



Now there is a non-causal path: $D \leftarrow X_2 \rightarrow Y$. We need to block it.

Example 2: Drug Efficacy with Confounding — SWIG Path Analysis

Try $\mathbf{S} = \{X_2\}$. Is $Y(d) \perp\!\!\!\perp D \mid X_2$?

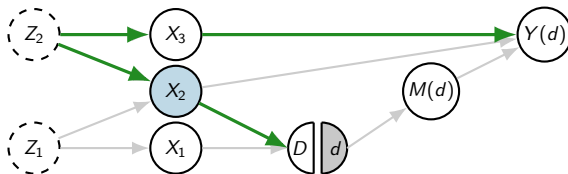


Path 1: $D \leftarrow X_2 \rightarrow Y(d)$

Blocked — fork at $X_2 \in \mathbf{S}$

Example 2: Drug Efficacy with Confounding — SWIG Path Analysis

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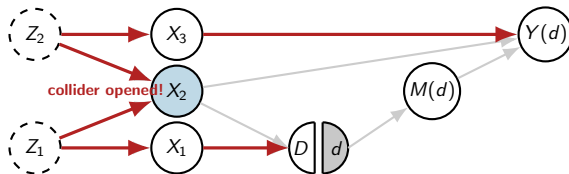


Path 2: $D \leftarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$

Blocked — chain at $X_2 \in \mathbf{S}$

Example 2: Drug Efficacy with Confounding — SWIG Path Analysis

Try $\mathbf{S} = \{X_2\}$. Is $Y(d) \perp\!\!\!\perp D \mid X_2$?



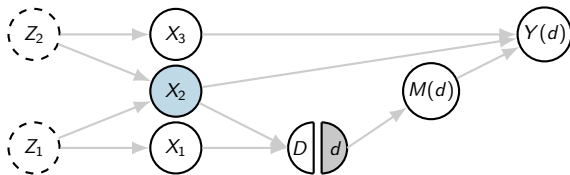
Path 3: $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$

X_2 is a **collider** on $Z_1 \rightarrow X_2 \leftarrow Z_2$. Conditioning on X_2 **opens** this collider!

Open!

Example 2: Drug Efficacy with Confounding — SWIG Path Analysis

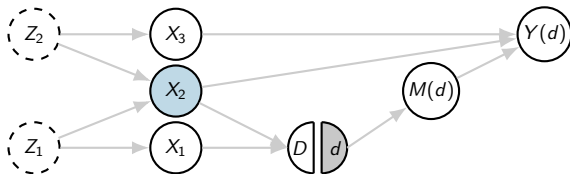
Try $\mathbf{S} = \{X_2\}$. Is $Y(d) \perp\!\!\!\perp D \mid X_2$?



Conclusion: $\mathbf{S} = \{X_2\}$ is **not sufficient**! Conditioning on X_2 blocks the fork/chain paths but **opens a collider path** through Z_1 and Z_2 .

Example 2: Drug Efficacy with Confounding — SWIG Path Analysis

Try $\mathbf{S} = \{X_2\}$. Is $Y(d) \perp\!\!\!\perp D \mid X_2$?



Valid Adjustment Sets

To block all paths, we need to also block the opened collider path:

- $\{X_2, X_1\}$ — blocks at X_1
- $\{X_2, X_3\}$ — blocks at X_3
- $\{X_1, X_2, X_3\}$ — blocks everywhere.

Do not condition on M (mediator)!

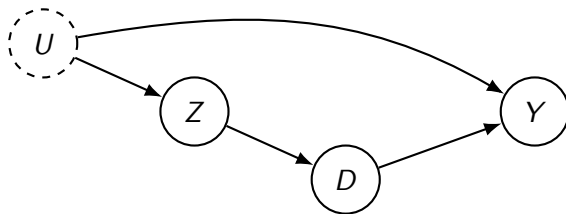
Example 3: Complete Proxy of Unobserved Confounder

Scenario: Estimating the effect of prenatal **multivitamin consumption** (D) on **birth defects** (Y).

- ▶ Unobserved **genetic factors** (U) affect both the likelihood of taking vitamins and the risk of birth defects.
- ▶ **Family medical history** (Z) fully mediates U 's effect on D : genetics influence family history, which then influences vitamin-taking behavior.

Task: Should we condition on Z (family medical history)?

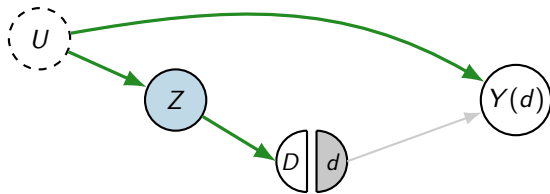
Example 3: Complete Proxy of Unobserved Confounder — DAG



Z is a **complete treatment proxy**: it captures all information flow from U to D .

Confounding path: $D \leftarrow Z \leftarrow U \rightarrow Y$.

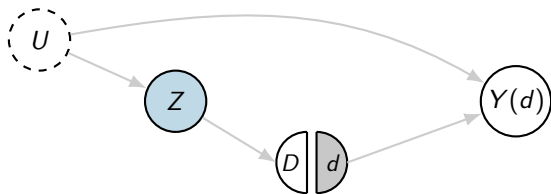
Example 3: Complete Proxy of Unobserved Confounder — SWIG



Path 1: $D \leftarrow Z \leftarrow U \rightarrow Y(d)$

Blocked — chain at $Z \in \mathbf{S}$

Example 3: Complete Proxy of Unobserved Confounder — SWIG



All Paths Blocked!

The only confounding path is blocked by conditioning on Z . $Y(d) \perp\!\!\!\perp D \mid Z$ holds.

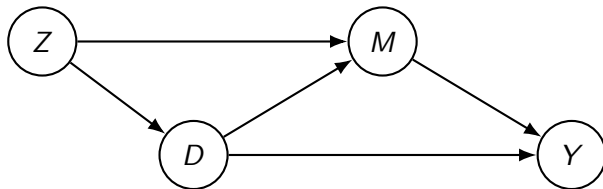
Example 4: Treatment-Mediator Confounder

Scenario: Estimating the effect of an **exercise program** (D) on **cardiovascular health** (Y).

- ▶ The program partly works through **diet changes** (M): exercise participants also improve their diet.
- ▶ **Socioeconomic status** (Z) is a common cause of both enrollment in the program and diet quality.

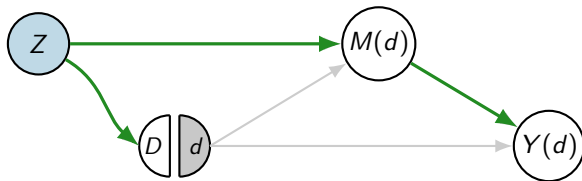
Task: Should we condition on Z (socioeconomic status)?

Example 4: Treatment-Mediator Confounder — DAG



Confounding path: $D \leftarrow Z \rightarrow M \rightarrow Y$.

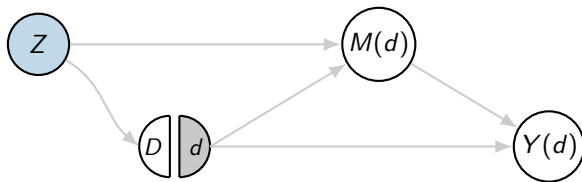
Example 4: Treatment-Mediator Confounder — SWIG Path Analysis



Path 1: $D \leftarrow Z \rightarrow M(d) \rightarrow Y(d)$

Blocked — fork at $Z \in \mathbf{S}$

Example 4: Treatment-Mediator Confounder — SWIG Path Analysis



All Paths Blocked!

Conditioning on Z blocks the confounding path. $Y(d) \perp\!\!\!\perp D \mid Z$ holds.

Example 5: Neutral Controls — Scenarios

5a: Outcome-Only Cause

Scenario: Effect of **tutoring** (D) on **test score** (Y). Student's IQ (Z) affects only Y , not who receives tutoring.

Question: Should we condition on Z ?

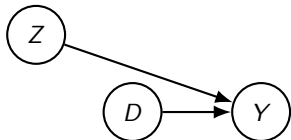
5b: Treatment-Only Cause

Scenario: Effect of **tutoring** (D) on **test score** (Y). Distance to tutor (Z) affects only who gets tutored, not the score directly.

Question: Should we condition on Z ?

Example 5: Neutral Controls — DAGs

5a: Outcome-Only Cause



No confounding path from D to Y .

Neutral / Good

No bias. Can **improve precision** by explaining variation in Y .

5b: Treatment-Only Cause



No confounding path from D to Y .

Neutral / Risky

No bias. But can **decrease precision** by explaining away useful variation in D .

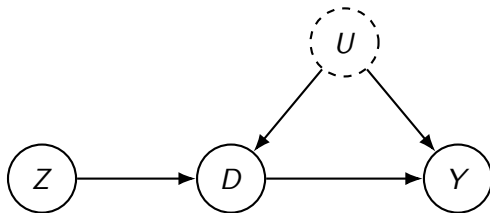
Example 6: Factor that only Affects a Confounded Treatment

Scenario: Estimating the effect of **military service** (D) on **lifetime earnings** (Y).

- ▶ A **draft lottery number** (Z) was randomly assigned and affects only whether someone serves.
- ▶ Unobserved **family background** (U) confounds both military service and earnings: wealthier families find ways to avoid service *and* their children earn more.

Task: Should we condition on Z (the lottery number)?

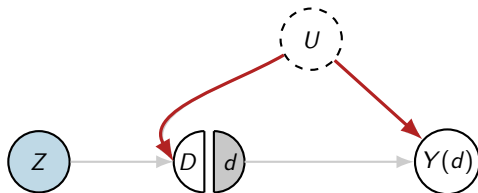
Example 6: Instrument — DAG



Z is an **instrument**: it affects Y only through D . The confounder U is unobserved.

Example 6: Instrument — SWIG Path Analysis

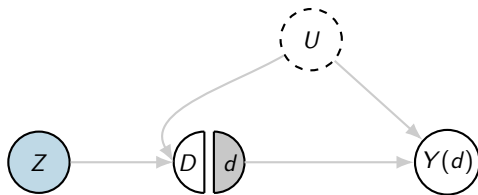
Conditioning on $\mathbf{S} = \{Z\}$:



Path: $D \leftarrow U \rightarrow Y(d)$ **Open!** — U is unobserved, and Z is not on this path so conditioning on it does nothing.

Example 6: Instrument — SWIG Path Analysis

Conditioning on $\mathbf{S} = \{Z\}$:



Bad control!

Conditioning on Z does **not** block the confounding path $D \leftarrow U \rightarrow Y(d)$. Worse: it removes the exogenous variation in D , leaving only the confounded variation driven by U . This **amplifies** the bias!

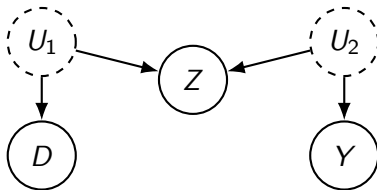
Example 7: Pre-Treatment Collider

Scenario: Estimating **peer effects** on civic engagement.

- ▶ D = engagement of friend 1, Y = engagement of friend 2.
- ▶ Z = whether they are friends (friendship status).
- ▶ Unobserved personality traits U_1 drive both friend 1's engagement and the friendship.
- ▶ Unobserved personality traits U_2 drive both friend 2's engagement and the friendship.
- ▶ This is **homophily**: similar people become friends.

Task: Should we condition on Z (friendship)?

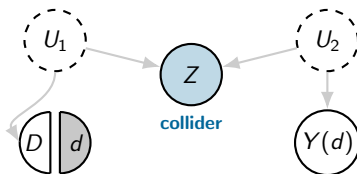
Example 7: Pre-Treatment Collider — DAG



Z is a **collider**: both U_1 and U_2 point into Z . The path $D \leftarrow U_1 \rightarrow Z \leftarrow U_2 \rightarrow Y$ is **blocked at the collider** without conditioning.

Example 7: Pre-Treatment Collider — SWIG Path Analysis

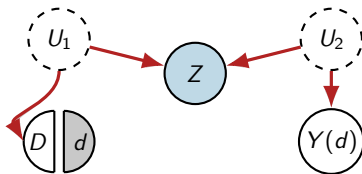
Conditioning on $\mathbf{S} = \{Z\}$:



Without conditioning: Path $D \leftarrow U_1 \rightarrow Z \leftarrow U_2 \rightarrow Y(d)$ is **blocked** at collider Z .

Example 7: Pre-Treatment Collider — SWIG Path Analysis

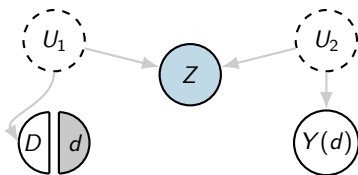
Conditioning on $\mathbf{S} = \{Z\}$:



With conditioning on Z : Path $D \leftarrow U_1 \rightarrow Z \leftarrow U_2 \rightarrow Y(d)$ is **opened!** Conditioning on a collider creates spurious association.

Example 7: Pre-Treatment Collider — SWIG Path Analysis

Conditioning on $\mathbf{S} = \{Z\}$:



Bad control! (M-Bias)

Z is a collider. Without conditioning, there is **no confounding** (D and Y are not connected by any open path). Conditioning on Z **opens** the path, creating spurious association!

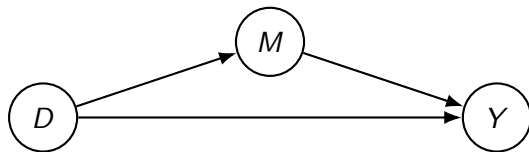
Example 8: Effect Mediator — Scenario

Scenario: Estimating the effect of a **job training program** (D) on **wages** (Y).

- ▶ The program works by improving **skills** (M), which then increase wages.
- ▶ M is a **mediator**: it lies on the causal path from D to Y .

Task: Should we condition on M (skills) when estimating the *total* effect of the program?

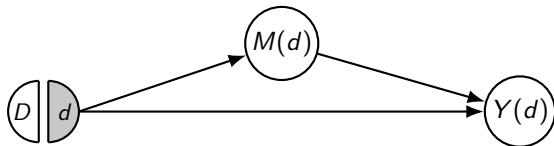
Example 8: Effect Mediator — DAG



M is a **mediator**: it lies on the causal path $D \rightarrow M \rightarrow Y$.

There are **no non-causal paths** from D to Y — no confounding at all!

Example 8: Effect Mediator — SWIG Analysis



The variable M does not appear on the SWIG!

We want to check $Y(d) \perp\!\!\!\perp D \mid M$, but M is a **post-treatment** variable. On the SWIG under $\text{fix}(D = d)$, only $M(d)$ appears — the interventional version. The original M is **not a variable** in this interventional world.

Since we cannot even **state** the conditional independence $Y(d) \perp\!\!\!\perp D \mid M$ using variables on the SWIG, we cannot use d-separation to verify it.

Example 8 Parenthesis: Direct Effect

Controlling for M estimates the “Controlled Direct Effect”

Controlling for M and applying identification by conditioning, we are estimating a form of a “direct effect”:

$$\text{Controlled-Direct-Effect (CDE)} = \mathbb{E}[Y(1, M) - Y(0, M)]$$

Interpretation. Intervene on both D and M , i.e., $\text{fix}(D = d, M = m)$. Look at differences when we flip the treatment variable, while mediator stays the same, i.e.,

$$\text{Individual-Direct-Effect}(m) = Y(1, m) - Y(0, m).$$

Then take the average of this direct effect, over the values of m , drawn from the non-interventional distribution of M :

$$\text{CDE} = \mathbb{E}[\text{Individual-Direct-Effect}(M)].$$

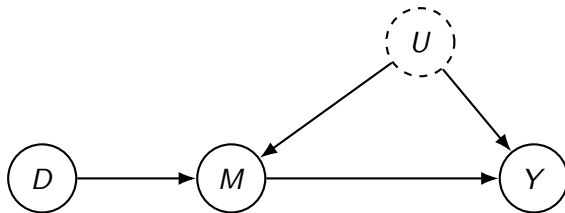
Example 9: Confounded Mediator

Scenario: Estimating the effect of a **new drug** (D) on **survival** (Y).

- ▶ The drug works through a **biomarker** (M): the drug changes the biomarker level, which affects survival.
- ▶ An unobserved **genetic factor** (U) affects both the biomarker and survival independently.

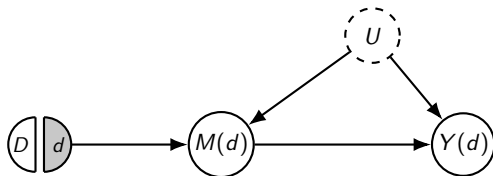
Task: Should we condition on M (the biomarker)?

Example 9: Confounded Mediator — DAG



M is a **confounded mediator**: it lies on the causal path $D \rightarrow M \rightarrow Y$, but U also affects both M and Y .

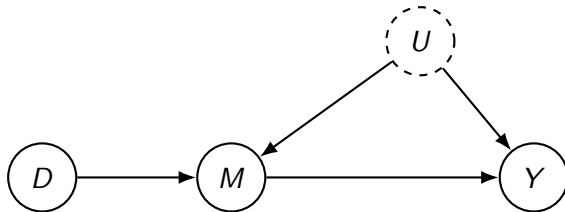
Example 9: Confounded Mediator — SWIG Analysis



The variable M does not appear on the SWIG!

We want to check $Y(d) \perp\!\!\!\perp D \mid Z$, but Z is **post-treatment**. On the SWIG, only $Z(d)$ appears. The original Z is **not a variable** in this interventional world, so we cannot use d-separation to verify the statement.

Example 9 Parenthesis: Not Even the Direct Effect



Controlling for confounded mediator *does not* estimate “Controlled Direct Effect”

Controlling for M we are opening a spurious correlation (confounding) path

$$D \rightarrow M \leftarrow U \rightarrow Y,$$

since M is a **collider** on that path.

Even if direct effect is zero (fully mediated effect via M), this method can estimate a non-zero effect, which captures correlation between M and Y through U .

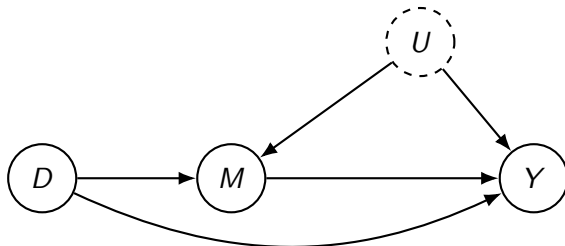
Example 10: Confounded Mediator — Scenario

Scenario: The **low birthweight paradox**.

- ▶ D = maternal **smoking** during pregnancy.
- ▶ Y = **infant mortality**.
- ▶ M = **low birthweight** (a post-treatment variable).
- ▶ U = unobserved **birth defects** (genetic abnormalities).
- ▶ Smoking causes low birthweight. Birth defects *also* cause low birthweight *and* infant mortality.
- ▶ Low birthweight also potentially causes mortality.
- ▶ Paradoxically, among low-birthweight babies, maternal smoking appears to be *protective*!

Task: Should we condition on M (birthweight)?

Example 10: Confounded Mediator — DAG



M is a **collider**: both D and U point into M ($D \rightarrow M \leftarrow U$). Without conditioning, the path $D \rightarrow M \leftarrow U \rightarrow Y$ is **blocked** at the collider. Conditioning on M would **open** this path, creating a spurious association!

Not even the direct effect

Conditioning on low birthweight does not estimate the controlled direct effect.

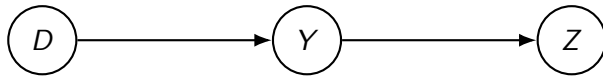
Example 12: Bad Control — Scenario

Scenario: The **Industrial Growth Puzzle**.

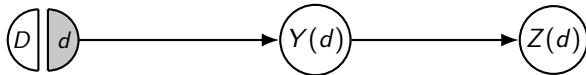
- ▶ D = **nutrition quality** in 19th century US/UK.
- ▶ Y = **height** of individuals.
- ▶ Z = **army enlistment** status.
- ▶ Better nutrition leads to taller people. But the data comes from army records — and taller men had more civilian opportunities, so were *less* likely to enlist.

Task: Is it valid to study this effect using only army enlistees (conditioning on Z)?

Example 12: Bad Control — DAG



Z is an **outcome of the outcome**: $D \rightarrow Y \rightarrow Z$.



Z does not even appear on the SWIG!

Key Lesson: Not All Controls Are Good Controls!

It matters what kind of node V is on a path:

Node Type	Structure	Conditioning
Collider	$\rightarrow V \leftarrow$	Opens the path
Common cause (fork)	$\leftarrow V \rightarrow$	Blocks the path
Mediator (chain)	$\rightarrow V \rightarrow$	Blocks the path

High-Level Rules of Thumb

- ▶ **Pre-treatment:** Adjust only for variables that are **ancestors** of D , Y , or both. Avoid colliders and instruments (in the presence of unobserved confounding).
- ▶ **Post-treatment:** Almost always a **bad idea** to adjust. Avoid mediators, colliders, and outcomes of the outcome.

Part IV: General Framework

DAGs, NPSEMs and SWIGs Beyond Examples

The General Framework: DAGs and Non-Parametric SEMs

Let us now state the general definitions beyond the examples.

Definition: Non-Parametric Structural Equation Model

A **non-parametric SEM** consists of:

- (i) A set of **endogenous variables** $V = \{V_1, \dots, V_p\}$ (observed or latent).
- (ii) A set of **exogenous noise terms** $\{\varepsilon_1, \dots, \varepsilon_p\}$, mutually independent.
- (iii) A set of **structural equations**: $V_i = f_i(\text{Pa}(V_i), \varepsilon_i)$, where $\text{Pa}(V_i) \subset V$ are the **parents** of V_i .

The parent structure must be **acyclic**: no variable is its own ancestor.

Definition: Directed Acyclic Graph (DAG)

The **DAG** associated with the SEM has one node per variable in V and a directed edge $V_j \rightarrow V_i$ whenever $V_j \in \text{Pa}(V_i)$. Acyclicity ensures **no directed cycles**.

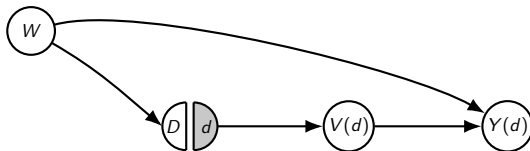
The General Framework: Interventions and SWIGs

Definition: Intervention $\text{fix}(D = d)$

Given a SEM and treatment $D \in V$, the **intervention** $\text{fix}(D = d)$ produces a new DGP: draw all exogenous noise from the **same distribution**; generate each V_i via f_i , **except** wherever D appears as a parent, replace it with d . Downstream variables become **interventional**: $V_i(d)$.

Definition: Single World Intervention Graph (SWIG)

The **SWIG** for $\text{fix}(D = d)$: **(1)** Split node D into two semi-circles — left half D (incoming) and right half d (shaded, outgoing). **(2)** Relabel downstream variables V_i as $V_i(d)$.



The General Framework: d-Separation

How do we formally check conditional independence in a DAG (or SWIG)?

Definition: d-Separation

Let $\mathcal{G}$ be a DAG and X, Y, Z disjoint node sets. X and Y are **d-separated** by Z ($X \perp_{\mathcal{G}} Y \mid Z$) if **every path** between X and Y is **blocked** by Z . A path is blocked if it contains a node V such that:

- (1) V is a **non-collider** ($\rightarrow V \rightarrow$ or $\leftarrow V \leftarrow$) and $V \in Z$; **or**
- (2) V is a **collider** ($\rightarrow V \leftarrow$) and neither V nor any descendant of V is in Z .

How to Use d-Separation for Causal Inference

- (1) Construct the **SWIG** for $\text{fix}(D = d)$.
- (2) Check whether D and $Y(d)$ are **d-separated** by a candidate set S .
- (3) If yes: $Y(d) \perp\!\!\!\perp D \mid S \Rightarrow$ conditional exogeneity holds and S is a **valid adjustment set**.

Takeaways

1. **d-Separation is the key tool:** To check conditional independence in a DAG or SWIG, enumerate all paths and verify each is blocked by the conditioning set.
2. **Three building blocks:** Every path is a sequence of chains, forks, and colliders. Conditioning blocks chains and forks, but **opens** colliders.
3. **Good vs. Bad Controls:** Not all pre-treatment variables are good controls — conditioning on a **collider** can introduce bias (M-bias). Post-treatment controls are almost always bad.
4. **The SWIG is a formal verification tool:** It turns an intuitive-but-vague concern (“maybe there’s confounding?”) into a precise, checkable statement. Always verify your identification strategy on the SWIG before estimating.

References

- ▶ Pearl, J. (2009). Causality. Cambridge university press.
- ▶ Richardson, T. S., & Robins, J. M. (2013). Single world intervention graphs (SWIGs): A unification of the counterfactual and graphical approaches to causality. Center for the Statistics and the Social Sciences, University of Washington Series. Working Paper, 128(30), 2013.
- ▶ Chernozhukov, V., Hansen, C., Kallus, N., Spindler, M., & Syrgkanis, V. (2026). *Applied Causal Inference Powered by ML and AI*. §7, §11.
- ▶ Cinelli, C., Forney, A., & Pearl, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, 53(3), 1071-1104.